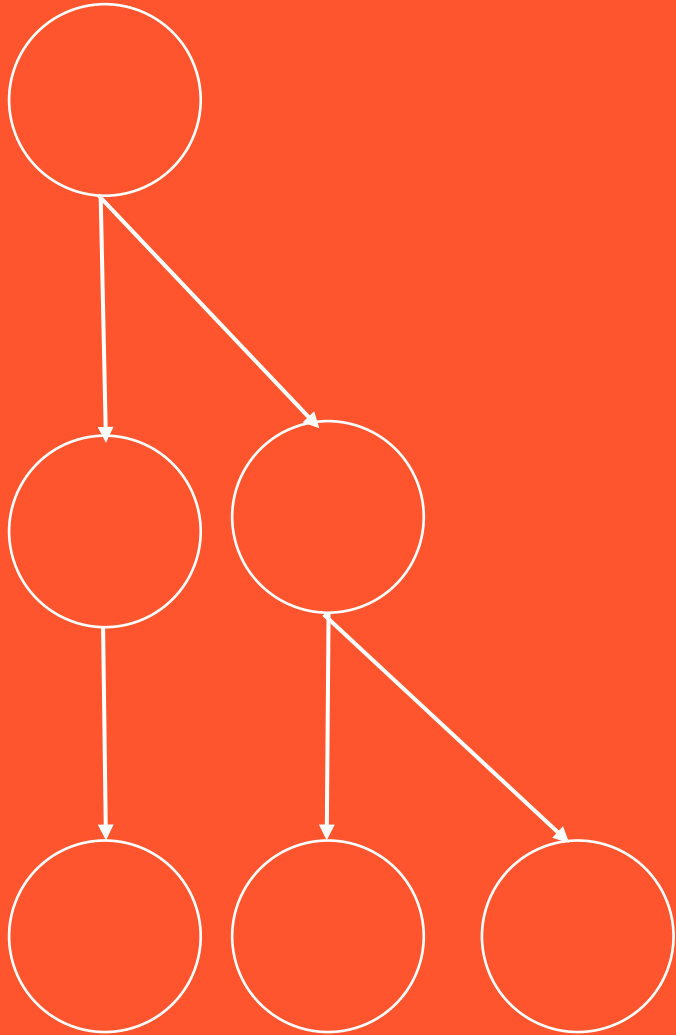


BCOG 100

Brains and Reinforcement Learning

Lecture 2: The Algorithmic Level



UNIVERSITY OF
ILLINOIS
URBANA-CHAMPAIGN

Reading Check 6-2

Password: actor

Question: Is social learning (learning from other people) classical or operant conditioning? Defend your answer.



Reinforcement Learning as an Algorithm



Computation

What does the system do?

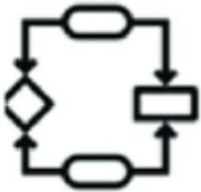
Recap & Overview

RL = linking actions to delayed outcomes

Today:

Today: algorithmic story across fields

Theme: from Hebb to error-driven learning



Algorithm

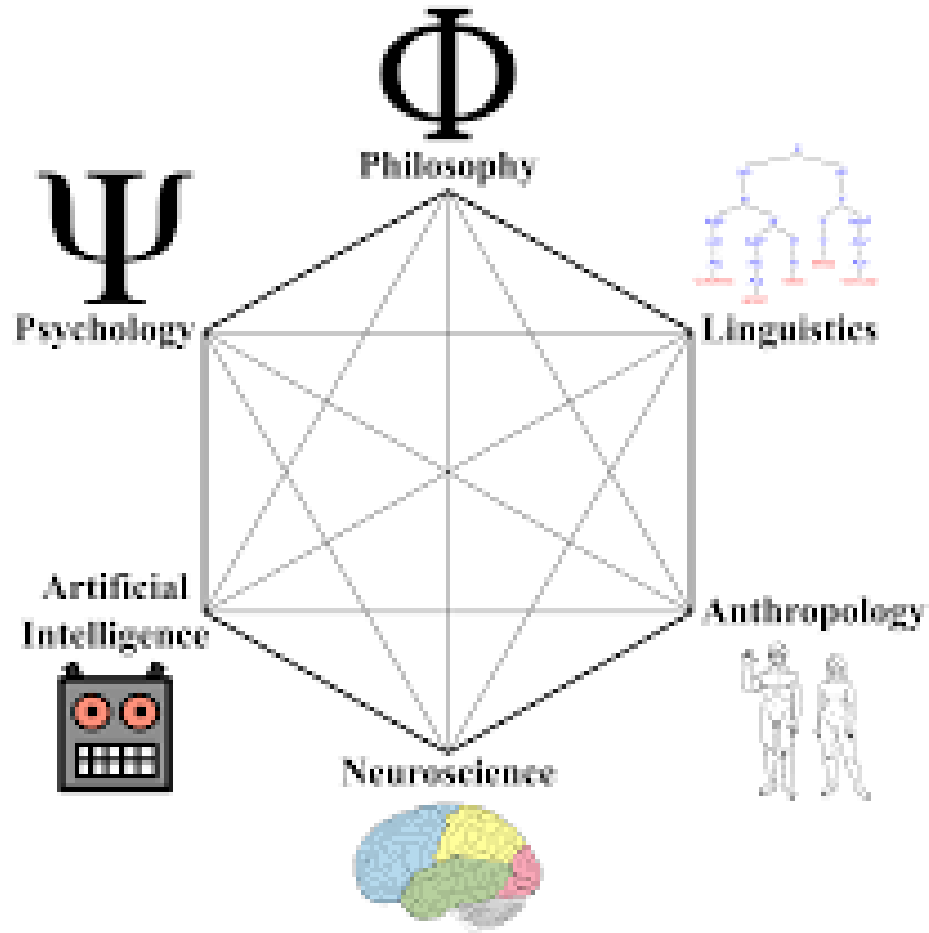
How does the system do it?



Implementation

How is the system realized?

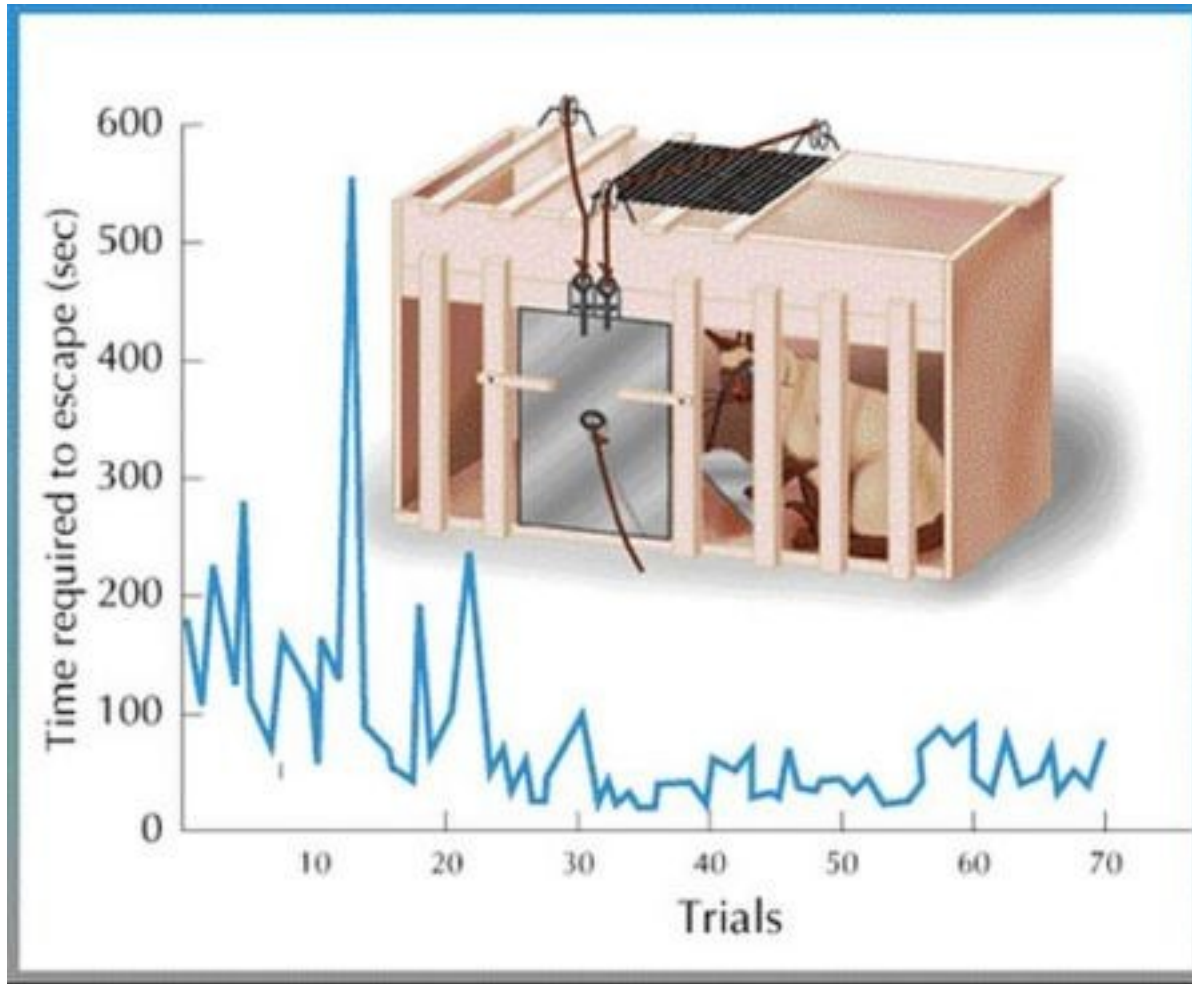
Reinforcement Learning as an Algorithm



A Multidisciplinary Story:

- Psychology: conditioning & trial-and-error
- Engineering: adaptive filters & delta rule
- AI: early neural nets & RL models

Reinforcement Learning as an Algorithm



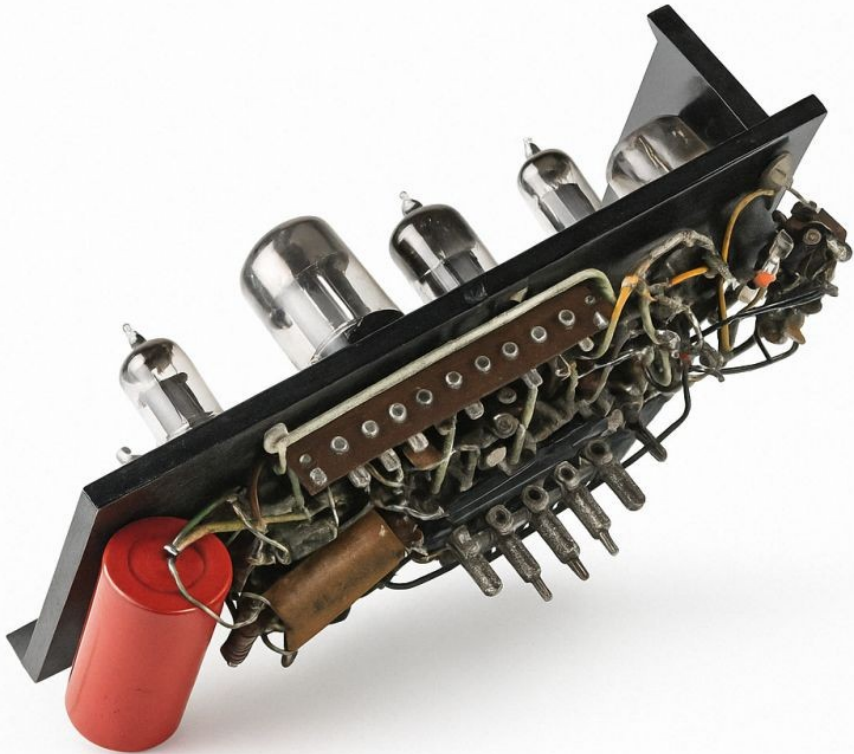
Thorndike's Trial-and-Error

- Cats in puzzle boxes
- Law of Effect: rewarded actions stamped in
- Early psychology of reinforcement

Reinforcement Learning as an Algorithm

1951 – ONE OF THE FIRST ARTIFICIAL NEURONS

Marvin Minsky's SNARC module



Minsky's SNARC (1951)

- Neural network maze-solver
- 40 neurons, reinforced recent synapses
- Worked in simple cases, failed in complex ones

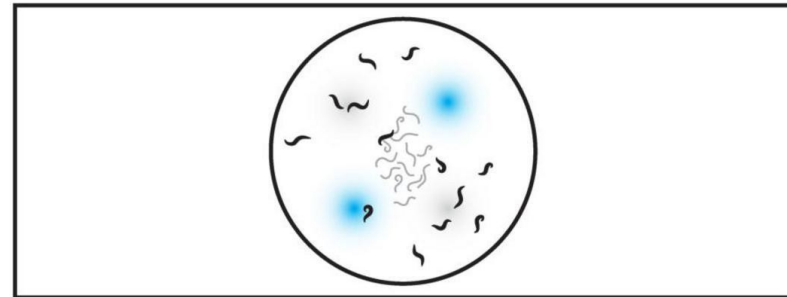
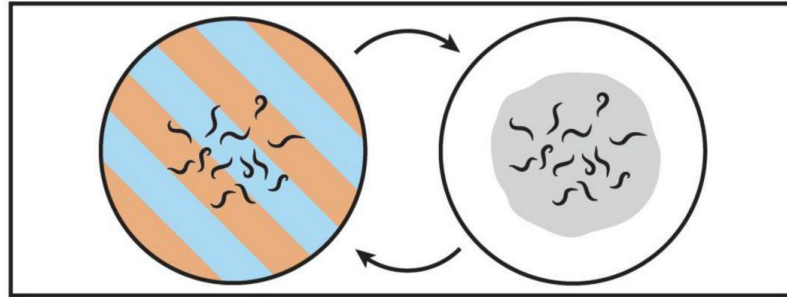
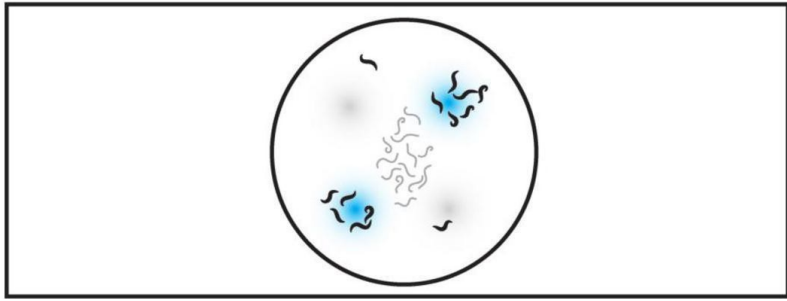
Reinforcement Learning as an Algorithm



The Temporal Credit Assignment Problem

- Which action deserves credit?
- Last action reinforced, earlier ones ignored
- Reinforce all actions equally → too slow

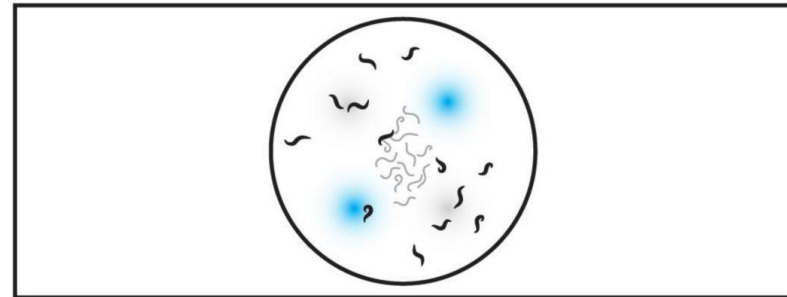
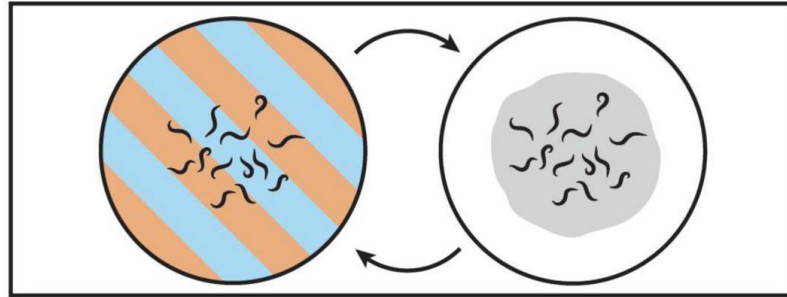
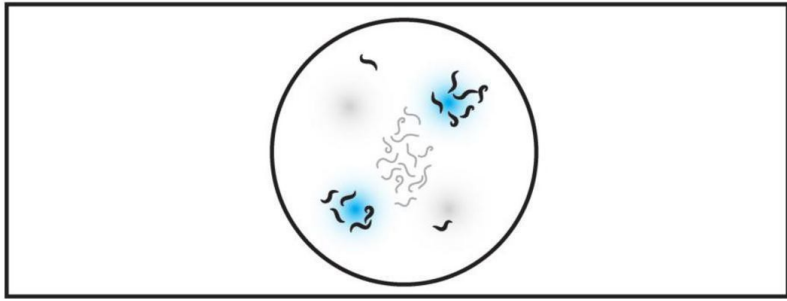
Reinforcement Learning as an Algorithm



How Hebb Can Support Reinforcement

- Associations shift action probabilities
- Cues paired with outcomes bias behavior
- Reinforcement can shape Hebbian plasticity

Reinforcement Learning as an Algorithm



Why Hebb Fails at RL

- Correlation-based, not outcome-based
- No way to handle delayed consequences
- Struggles with credit assignment

Reinforcement Learning as an Algorithm

Rescorla–Wagner model (1972)

$$\Delta V_A = \alpha \beta (\lambda - \Sigma V_T)$$

ΔV_A = change in associative strength towards stimulus A

α = salience of the CS

β = speed of learning for the given US

λ = theoretical asymptote of learning. It is a reflection of the US.

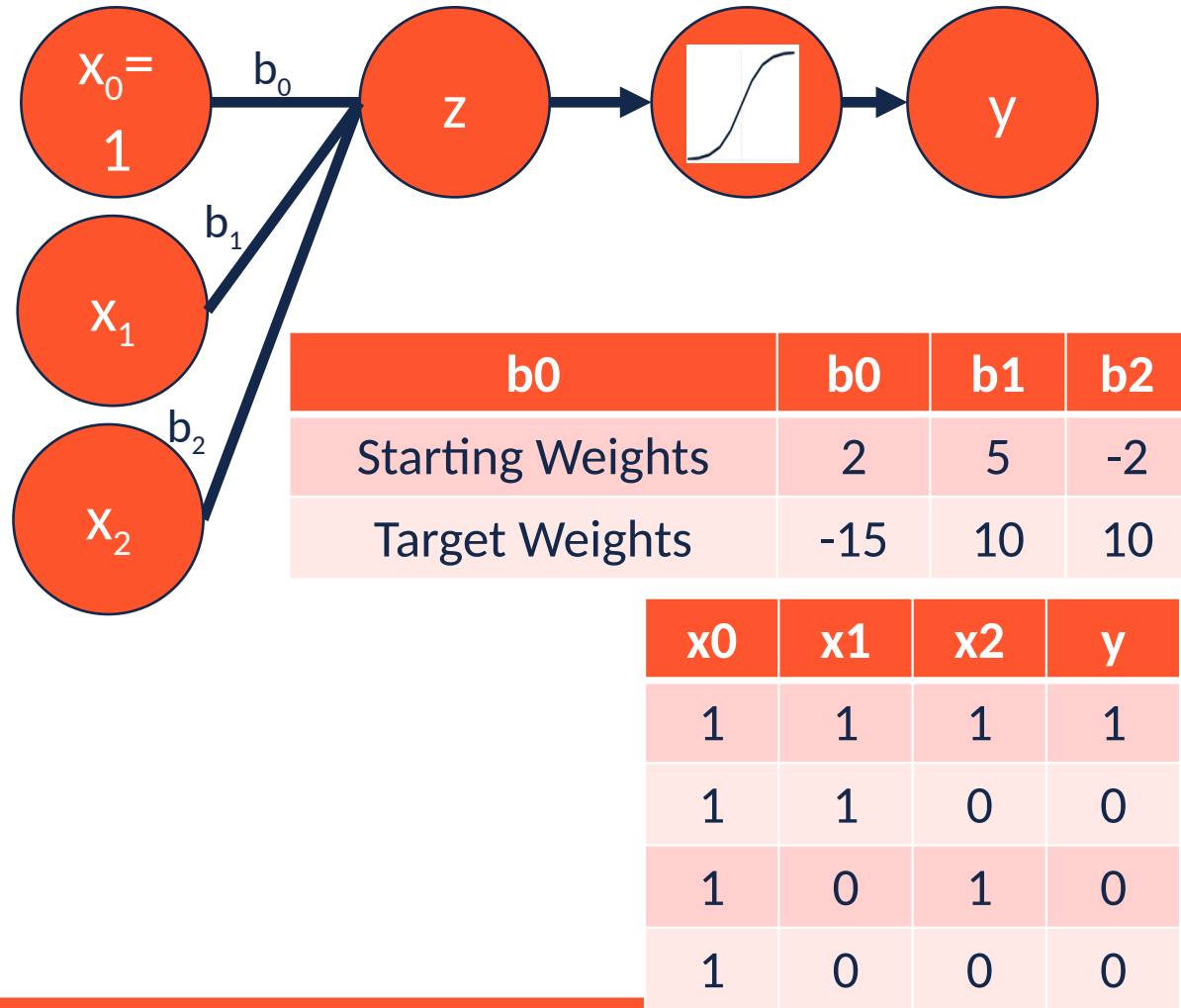
ΣV_T = cumulative associated strength

Error-Driven Learning Emerges

- Widrow–Hoff delta rule (1960)
- Rescorla–Wagner model (1972)
- Independent origins, same core idea



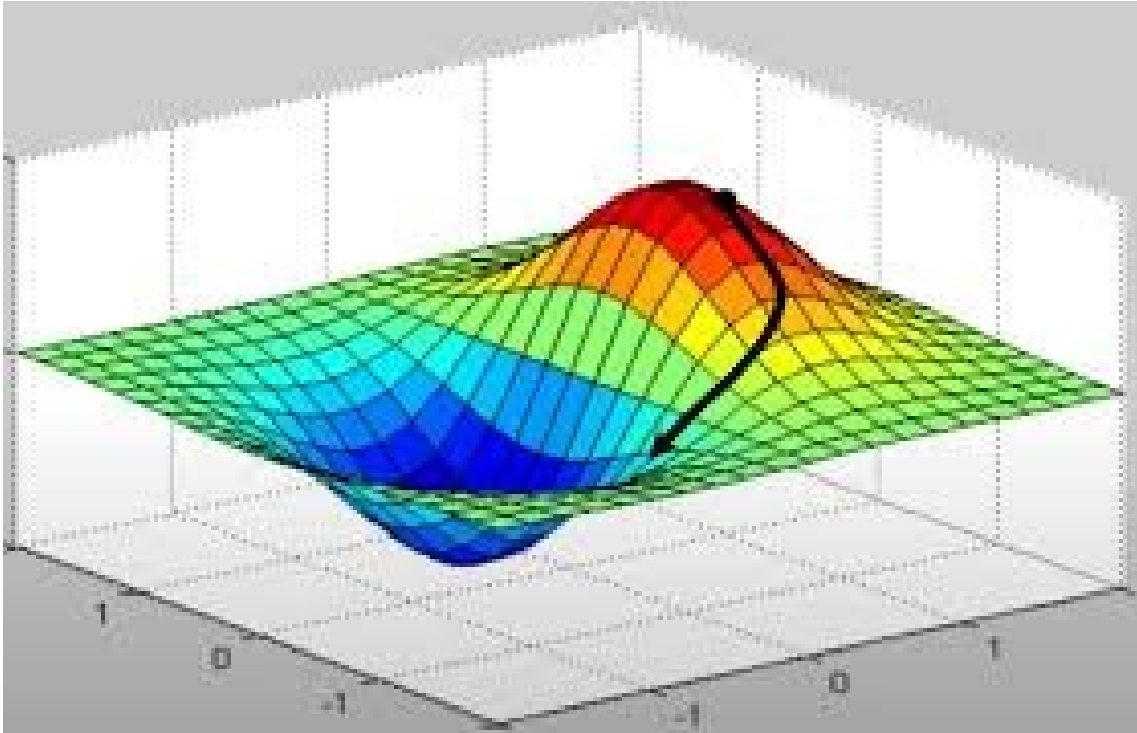
Reinforcement Learning as an Algorithm



Worked Example: Learning AND

- Task: $(1,1) \rightarrow 1$, all others $\rightarrow 0$
- Hebb: strengthens co-occurrence
- Delta: adjusts by error correction

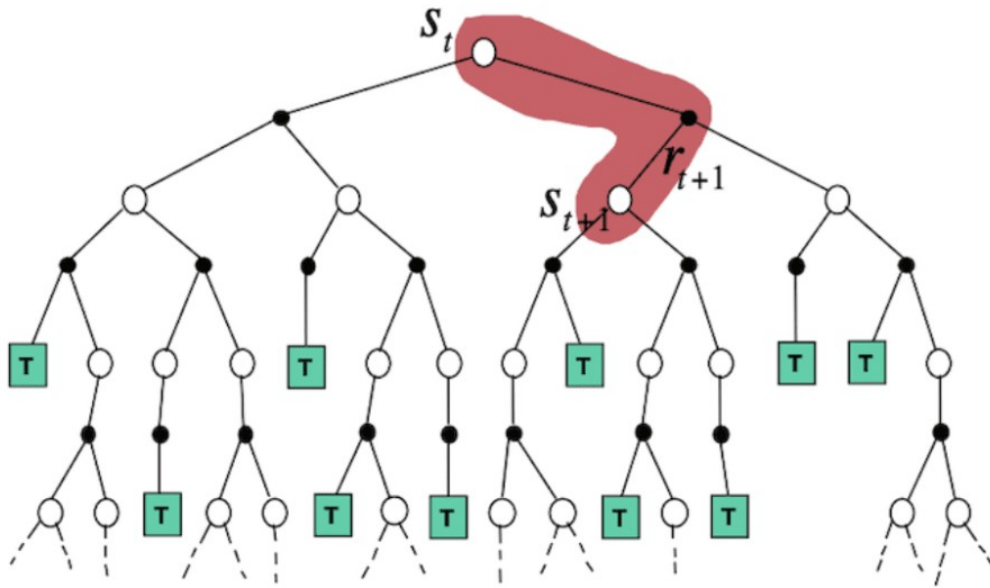
Reinforcement Learning as an Algorithm



From Delta Rule to RL

- Error signals stronger than correlation
- Psychology + engineering converge
- Foundation for modern RL

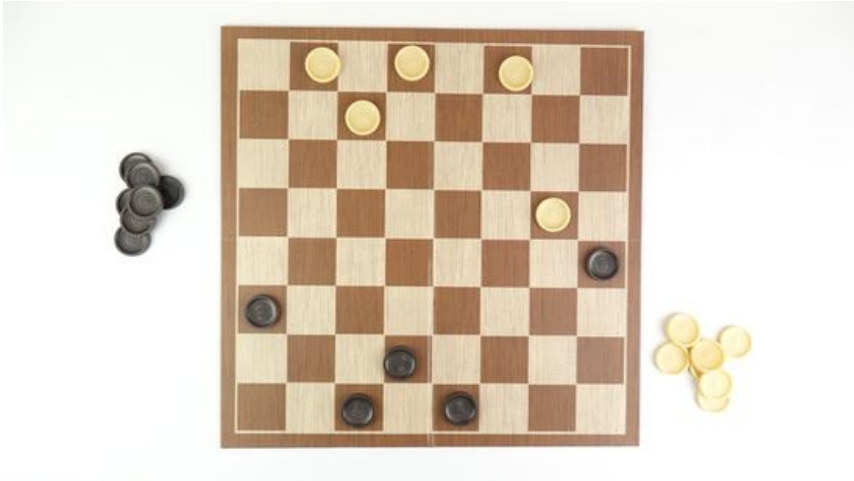
Reinforcement Learning as an Algorithm



Sutton's Breakthrough: Temporal Difference Learning

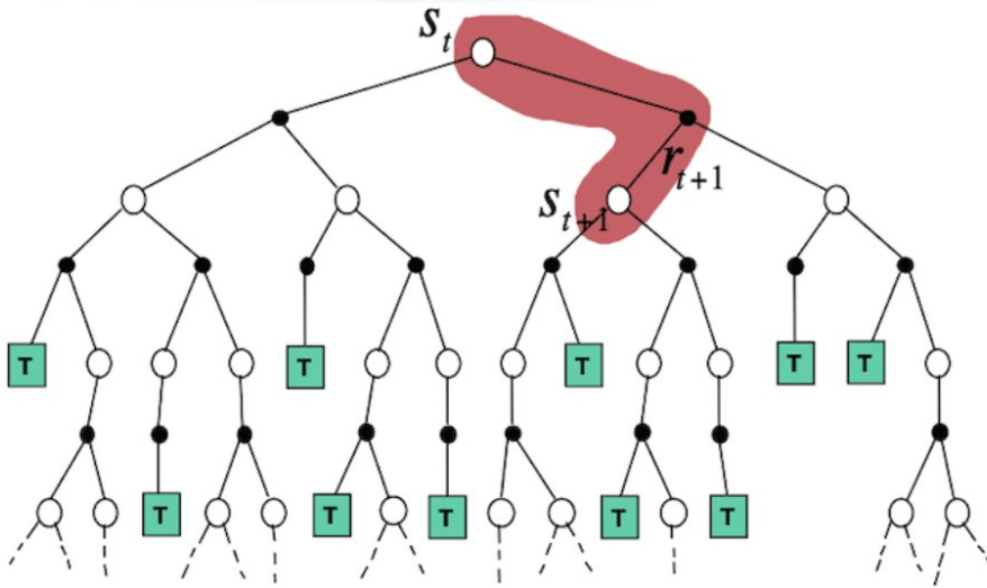
- Update predictions after each action
- Error = new prediction – old prediction
- Helps choose better actions

Reinforcement Learning as an Algorithm

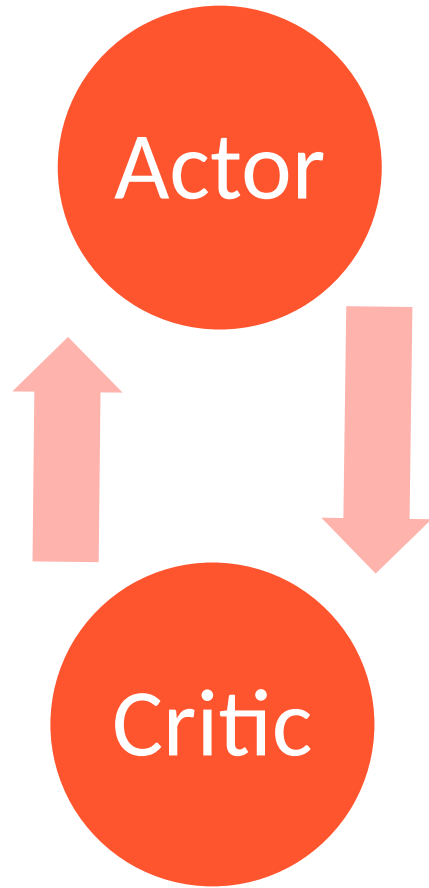


Sutton's Breakthrough: Temporal Difference Learning

- Update odds of winning after each move
- Error = new prediction – old prediction
- Helps choose better actions



Reinforcement Learning as an Algorithm



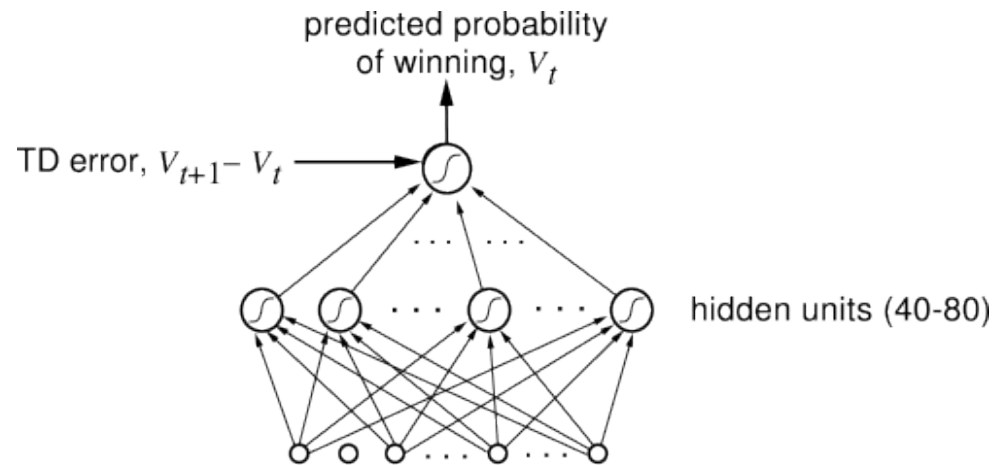
Two Roles in Temporal Difference Learning

- Evaluating positions (predictions)
- Choosing moves (actions)
- Both are needed in checkers

Reinforcement Learning as an Algorithm

Starting position :

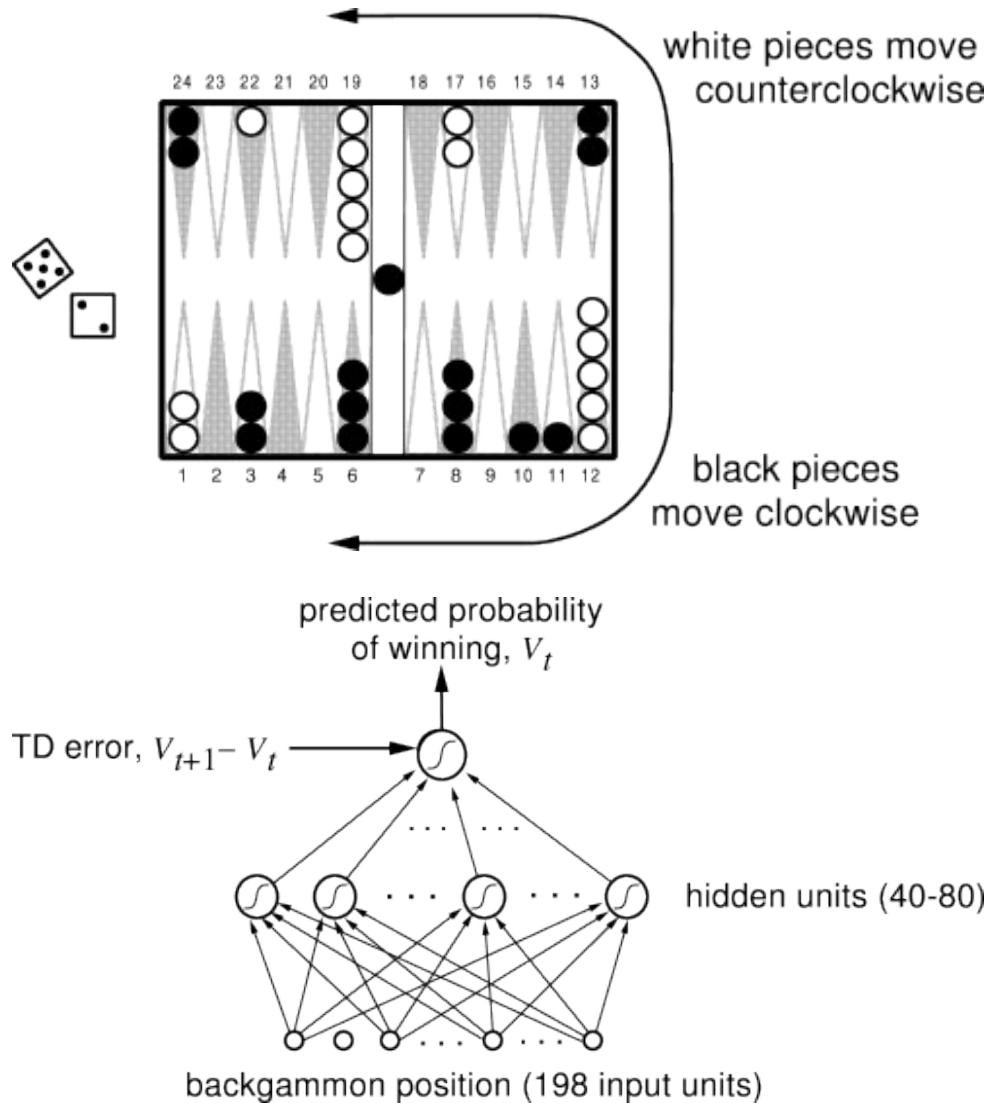
	0	0	0	0
0	0	0		
	0			0
			0	
			X	X
	X	X	X	X
X	X	X	X	



Actor-Critic Model

- **Critic:** evaluates states (predicted value)
- **Actor:** chooses actions based on evaluations
- TD error links them together

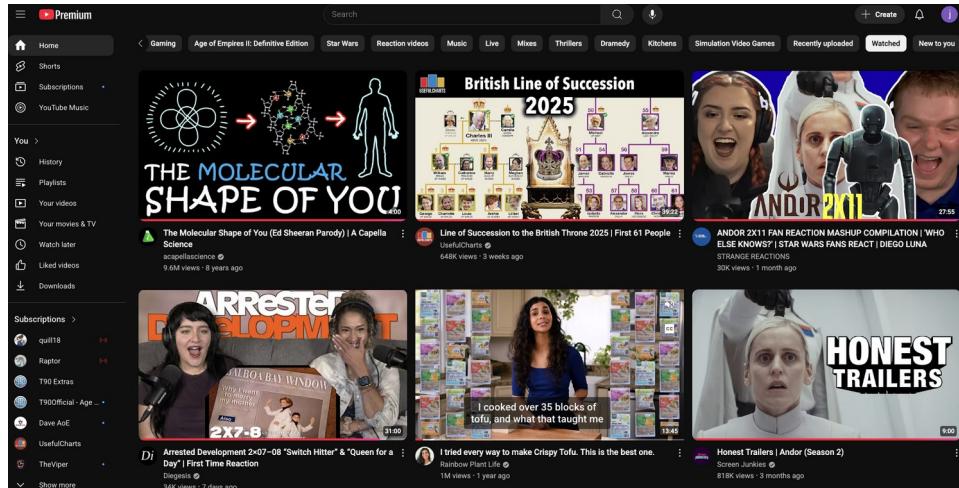
Reinforcement Learning as an Algorithm



TD Learning in Practice: TD-Gammon

- Tesauro's backgammon AI (1990s)
- Trained by self-play with TD updates
- Reached near-expert human level

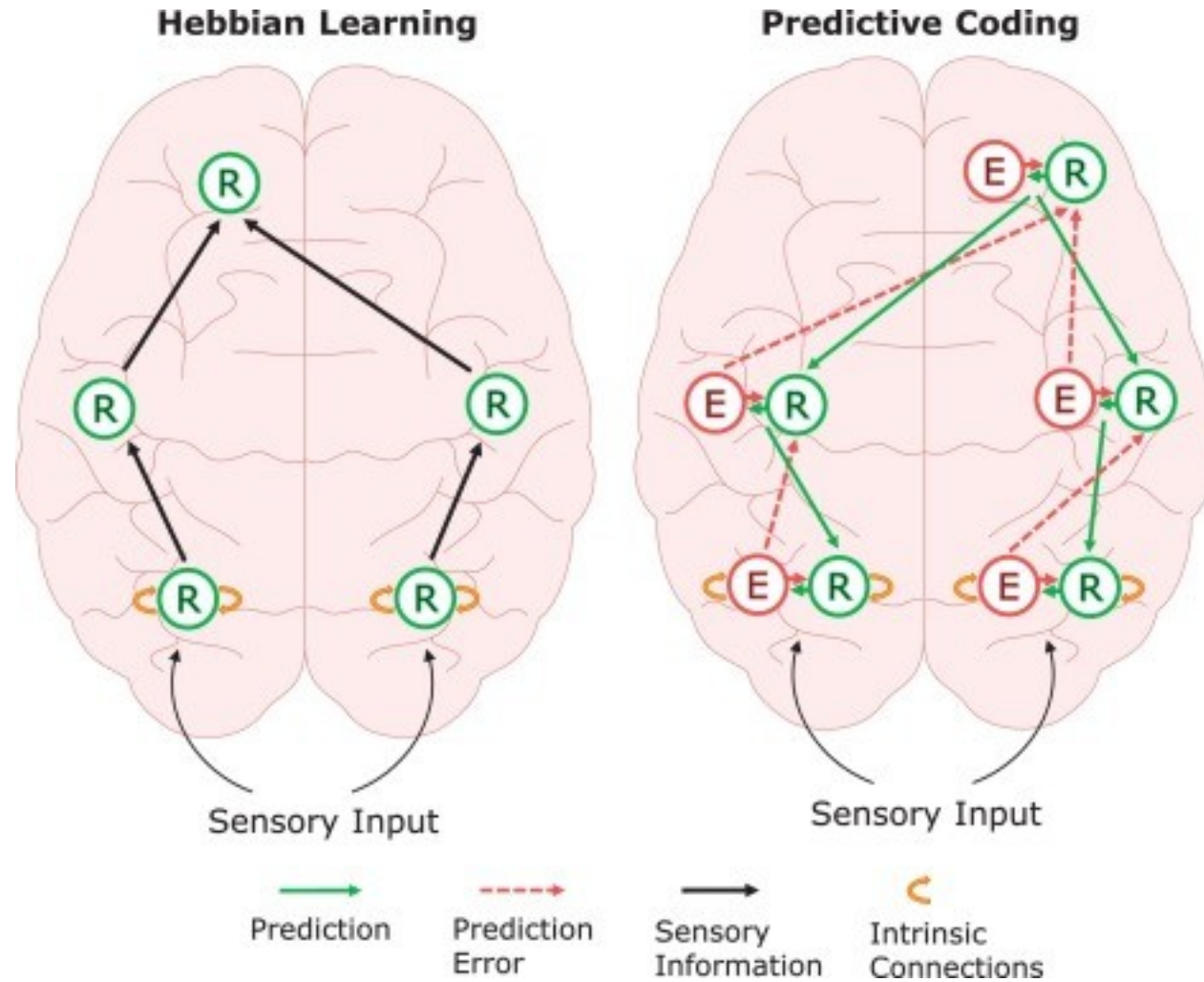
Reinforcement Learning as an Algorithm



RL Today: Ubiquitous Applications

- Game-playing AIs (Chess, Go, StarCraft)
- Robotics & control systems
- Everyday tech: ads, recommendations, resource allocation

Reinforcement Learning as an Algorithm



Algorithmic Takeaways

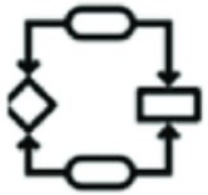
- Hebb: correlation-based associations
- Error-driven: prediction error updates
- Convergence across psychology, engineering, AI

Reinforcement Learning as an Algorithm



Computation

What does the system do?



Algorithm

How does the system do it?



Implementation

How is the system realized?

What We Learned / What's Next

- RL needs prediction error, not just Hebb
- Delta rule → TD learning → actor-critic
- Next: how the brain implements these signals